Hao Lun Li

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EDUCATION

University of British Columbia, Vancouver — Sep 2025 - Apr 2029
Bachelor of Science in Computer Science, Co-op Program, Dean's List — GPA: 88.3%

TECHNICAL SKILLS

- **Languages**: Java, Python, TypeScript/JavaScript, C++, R, HTML/CSS/JSON
- **Frameworks & Tools**: React, Expo, Vercel, Git, Junit, Tidyverse, GitHub Actions, Rest APIs
- **Cloud & Databases**: AWS (EC2, S3, Aurora & RDS), Supabase (Auth, Tables), PostgreSQL

PERSONAL PROJECTS

actually (IOU Tracker) | Typescript, React Native, Expo, Vercel, Supabase | Live — Apr 2026 - Present

- Built and launched a full-stack cross-platform (iOS, Android, web) IOU tracker app for friends in **React Native** and Typescript, deployed on **Vercel** with EAS build profiles for Play Store and soon App Store release.
- Designed a PostgreSQL schema on **Supabase** with Row Level Security, using triggers to enforce two-sided settlement and security-definer functions for account linking, verified by a 48-check SQL behaviour test.
- Built account-free guest access via a Deno **Edge Function** with SHA-256-hashed 128-bit share-link tokens, owner-scoped service-role queries, and Postgres-backed per-IP and per-link rate limiting.

calcli (Google Calendar CLI) | C++, CMake, vcpkg, Catch2 | GitHub — Sep 2026 – Present

- Built a **C++** command-line tool that creates Google Calendar events from natural-language input (e.g. calcli add "Exam" Friday 2:30 1 hour), with a unit-tested parser for weekdays, clock times, and durations.
- Implemented the full OAuth 2.0 flow (consent, token exchange, caching, refresh) and a **REST** client over libcurl for the Google Calendar API, with RFC 3339 JSON serialization via nlohmann::json.
- Shipped v0.2.0-alpha as a standalone statically linked Windows binary with a **CI/CD pipeline** using GitHub Actions that runs the full **Catch2** test suite and publishes releases on version tags.

TECHNICAL EXTRACURRICULAR ACTIVITIES

Team SHIJ, Amazon Robotics Hackathon 2026 | Git, Python | GitHub — Sep 2026

- Designed and implemented a Djikstra-based dynamic routing algorithm in **Python** for warehouse drive units in a team of 4, caching reverse-graph shortest-path distances so units re-evaluate the optimal next hop every timestep.
- Built congestion and deadlock-aware coordination logic to prevent race conditions between idle units as well as integrated detour-vs-wait cost comparisons using live data to achieve **100% pod delivery** (vs. 66.7% for a baseline greedy algorithm) and completing all test scenarios in ~75% fewer time steps.

Generic Health Consulting, WiDS ASA Datafest 2026 | R, Python | GitHub — Mar 2026

- Analyzed over **340 000** lines of distributed medical history information from Stormont Vail Health in **R** to accurately locate and evaluate misallocated post-operative attention on varying patient needs.
- Classified patients into 4 archetypes through **KNN-Classification** providing Stormont Vail Health with better triage tools improving patient recovery and lowering hospital operating costs via more effective prioritization.

WORK EXPERIENCE

Aquatic Leader, City of Coquitlam — Feb 2024 - Present

- Communicated with and lifeguarded up to **500** patrons daily, addressing concerns, answering questions, and resolving issues efficiently in a fast-paced environment.